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Review article

Smartphone as a monitoring tool for bipolar disorder: a systematic review including data analysis, machine learning algorithms and predictive modelling

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ABSTRACT

Background: Bipolar disorder (BD) is a chronic illness with a high recurrence rate. Smartphones can be a useful tool for detecting prodromal symptoms of episode recurrence (through real-time monitoring) and providing options for early intervention between outpatient visits.
Aims: The aim of this systematic review is to overview and discuss the studies on the smartphone-based systems that monitor or detect the phase change in BD. We also discuss the challenges concerning predictive modelling.
Methods: Published studies were identified through searching the electronic databases. Predictive attributes reflecting illness activity were evaluated including data from patients' self-assessment ratings and objectively measured data collected via smartphone. Articles were reviewed according to PRISMA guidelines.
Results: Objective data automatically collected using smartphones (voice data from phone calls and smartphone-usage data reflecting social and physical activities) are valid markers of a mood state. The articles surveyed reported accuracies in the range of 67% to 97% in predicting mood status. Various machine learning approaches have been analyzed, however, there is no clear evidence about the superiority of any of the approach.
Conclusions: The management of BD could be significantly improved by monitoring of illness activity via smartphone.

1. Introduction

Bipolar disorder (BD) is a chronic, recurrent illness with a high morbidity and mortality rate [1]. The prevalence in the general population is estimated to be around 2% [2]. In the course of disease, there are fluctuations between different mood phases, ranging from depression to manic episodes, as well as mixed states [3]. The recurrent and chronic nature of bipolar disorder results in a vast impact on psychosocial functioning and quality of life [1], as well as a high burden of this disease [4] and high socio-economical costs [5]. The risk of a new episode can be reduced significantly by the treatment with mood stabilizing drugs [6]. Early pharmacological intervention is crucial as it reduces the conversion rates to full-blown illness and decreases the severity of symptoms [7]. However, patients often show lack of insight about their symptoms and are not aware of the need of treatment (this especially concerns manic phases) [8]. Also, the current treatment paradigm for bipolar disorder involves routine monitoring of patients through regular clinical visits [9], but the frequency of control visits are often insufficient [10]. Consequently, adjustment of the treatment often takes place late, sometimes during a full episode [7]. Hence, there is a need for more effective strategies of preventing affective episodes [11]. One of the possible options could be a continuous monitoring of various parameters reflecting illness course such as mood, sleep or activity via smartphone.

In recent years, there is an increasing growth of interest in the utility of new technologies, such as electronic platforms or smartphone applications, for psychiatric care [12]. Specialized smartphone applications are being introduced to various psychiatric disorders, including

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bipolar disorder [13], postnatal depression [14], addiction [15] and caregivers of psychiatric patients [16]. Mobile phones offer unique capabilities for monitoring of depressive and manic symptoms through the real-time self-reported data, as well as through the data automatically collected by smartphone such as speech activities or social and physical activities [17,18]. In the course of the illness, there are significant disturbances especially in diurnal rhythms and daily life regularity [79]. Therefore, the management of the illness could be focused on self-tracking the behavioural and sleep-wake patterns. Initially, traditional paper-and-pencil daily mood charts were applied, but modern technology - smartphones offer even a greater opportunity for continuous assessment of these variables. It has been demonstrated, that the rates of compliance and psychometric properties are significantly better for mobile versions as compared to paper-and-pencil mood ratings [19]. Moreover, it was demonstrated that patients prefer mobile-based tracking over paper-based methods [20,21].

The central features of BD are also the abnormalities in psychomotor and social activities, typically with psychomotor retardation, social withdrawal and paucity of speech during depression and increased motor, social and speech activity during mania. The changes in the manner of speaking reflect the mood state very accurately and it is used intuitively by clinicians in everyday practise. Reduced speech activity, changes in voice features and pause related measures may be a sensitive marker of prodromal symptoms of depression or effect of treatment [22–24]. On the other hand, an increased speech activity may predict a switch to hypomania [25]. Some of the promising features, such as jitter, formants, spectral slope or delta-mel-cepstrum, originate from speech processing and have rather technical character [26]. Therefore, the interest in the potential offered by smartphones in combination with machine learning approaches to prediction of phase change in BD is constantly growing and there has been a great deal of work on this subject in recent years.

Several reviews have been published recently on this topic. Faurholt-Jepsen et al. [27] acknowledge the transformative potential of mobile and wireless technologies for monitoring of BD patients, but conclude that it is still too early to choose the combination of smartphone data that are the best predictors for the levels of depressive or manic symptoms. Some reviews have more specific focus. Rajagopalan et al. [28] describe the potential benefits of digital platforms and point out poor evidence base and data security issues. Daus et al. [29] assess the patients' willingness to have their health monitored with on-line tools and find that BD patients are generally very positive towards these novel techniques. Nicholas et al. [13] reviewed BD-related apps already available in the Google Play and iOS stores assessing their features and the quality of their content. They concluded that these apps were not in line with evidence-based medicine. Cummins et al. [26] survey both technical and medical aspects of acoustic analysis of speech for depression diagnosis.

The distinguishing features of this review are that: 1) the particular emphasis is placed on the BD phase change detection and related opportunities and challenges; 2) we investigate the data analysis and machine learning potential for the field; and 3) this review covers a broad set of systems and applications.

1.1. Objective

The aim of this paper is to overview and discuss the studies on the smartphone-based systems that monitor or detect the phase change using either subjective ratings or objectively measured data reflecting illness activity in patients with BD. We have focused on monitoring to explore how the real-time data and digital phenotyping can enhance the care between outpatient visits. The specific objectives of this study were to examined current literature in terms of: 1) the association of self-monitored data with mood state; 2) the association of objective behavioural data with mood state; 3) the association of voice features with mood state; 4) data analysis and machine learning algorithms applied in this field.

2. Methodology

Published studies were identified through searching the electronic databases in March 2019: MEDLINE, PsychINFO and Embase for relevant articles. Follow-up citations of retrieved studies were scanned for other relevant studies. Machine-learning related studies were searched with a query to the Scopus database in May 2019. The following terms/key words were employed: "bipolar" or "BD" combined with one of the following "smartphone", "mobile", or "phone". We limited the results to papers published after the year 2000.

Articles were reviewed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Two independent raters screened the abstracts of the resultant de-duplicated and not English language papers. During the further screening of abstracts by two raters only studies about mHealth and concerning subject were included for further review. The two raters reviewed the full articles. The full text of 37 papers were included for qualitative review. Papers not presenting empirical studies (articles that were systematic reviews) and study protocols were excluded. The two independent raters compared findings to achieve consensus and discrepancies were resolved by a third rater until a final list of studies pertaining to the evaluation was compiled. The study selection process is illustrated in Fig. 1

3. Results

The initial query search results in MEDLINE, PsychINFO, Embase and Scopus databases resulted in 2422 records. Reference lists searching resulted in n = 15 additional publications. After removing duplications and texts published in language other than English we included n = 202 abstracts of papers published in 2000-2019 into further proceedings. After the scan of the titles and abstracts, 165 further articles were excluded (not concerning bipolar disorder, not Health, not concerning subject: smartphone-based systems that monitor or detect the phase change in BD), thereby resulting in 37 papers. The full-text reading resulted in the exclusion of n = 10 additional articles (not empirical studies, study protocols). The n = 27 articles in the final set were read and analyzed. Most active research clusters that have published on this topic are presented in Fig. 2.

3.1. Self-assessment data

We have identified 12 papers investigating the effects of self-monitoring apps. Table 1 contains the detailed characteristics.

For instance, MONARCA app offers a possibility to assess, log and monitor mood, activity, hours of sleep, anxiety level, and medication use, which are self-reported and can be shared with clinician [30,31[75]]. There are also apps providing personalized psychoeducation program or mobile therapy (for instance SIMPLe app, [32]). The SIMPLe app showed efficacy at improving sleep, social rhytms and eating pattern [33].

Faurholt-Jepsen et al.[17,30,34]) have shown, that self-monitoring correlates significantly with scores on clinical rating scales assessing depression (Hamilton Depression Rating Scale - HDRS) and mania (Young Mania Rating Scale - YMRS) severity. Although patients are generally able to ascertain the depressive symptoms, self-identification of hypomanic/manic symptoms is less reliable due to decreased illness insight [34]. However, it is not proven that daily self-monitoring could predict the phase change or reduce depressive or manic symptoms [30].

3.2. Smartphone-usage data

Table 2 presents the detailed characteristics of 14 studies investigating the relations between the mood state and the objective data

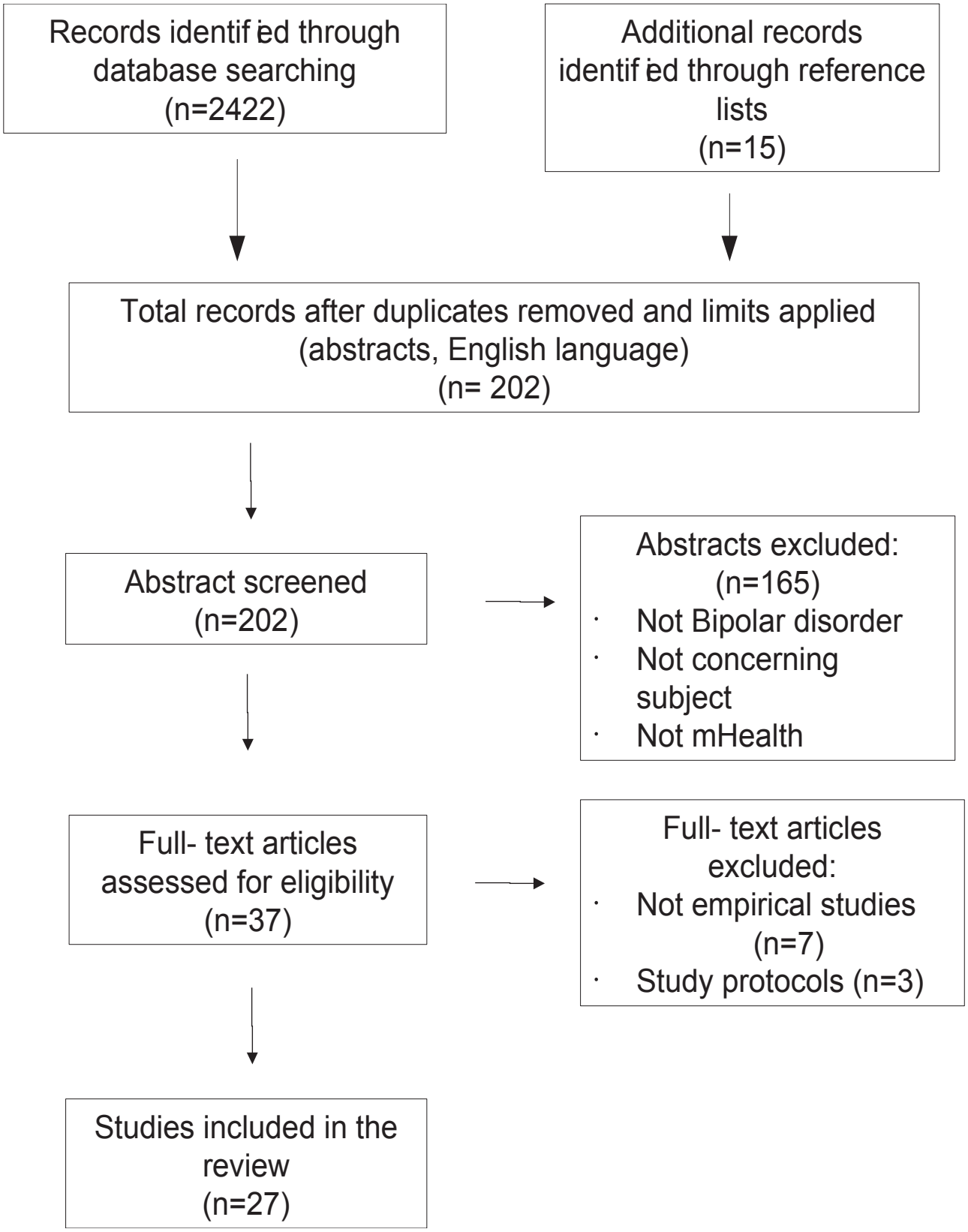

Fig. 1. Flow chart: an overview of the study selection process.

collected via smartphone among BD patients.

The following objective features were extracted from the smartphone-usage data in different studies: phone call statistics, phone usage, physical activity (measured through the accelerometer), mobility (based on satellite navigation), physiological data (e.g. heart rate variability). Several lines of evidence indicate, that these objective data correlate with the scores on HDRS and YMRS scales [30,34,35]. Osmani et al. [36] and [17,30]) showed a correlation between objective clinical

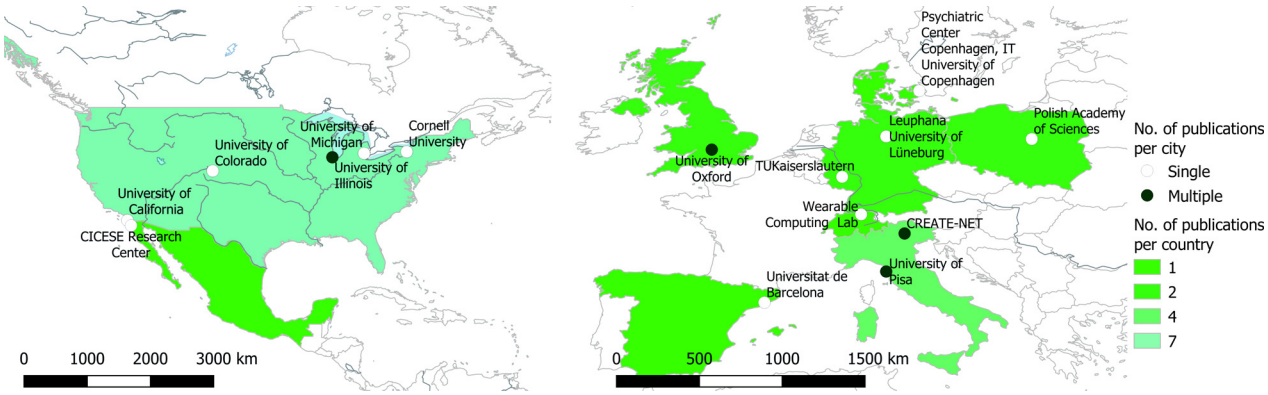

Fig. 2. The geographical location of the most active research clusters that have published in the field of smartphone-based monitoring in bipolar disorder.

assessment of mood state and the physical and social activity (reflected by the number and duration of calls and text messages). Beiwinkel et al. [37] demonstrated that an increase in depression severity was predicted by a decline in social and physical activity. An increase in mania severity was predicted by an decrease in physical activity and higher social communication. Similarly, Grünerbl et al. [35] conducted a study based on smartphone-sensing, in order to classify the state and detect state changes. In this study, social interaction, physical motion and travel patterns were assessed. In turn, Zulueta et al. [38] stated, that mood phases in bipolar disorder correlate with specific changes in mobile phone usage (passively collected mobile phone keyboard meta data).

3.3. Voice data

Smartphones provide a unique opportunity for monitoring the speech in a naturalistic settings. We have found 7 studies (Table 3) assessing the mood state using voice features. During phone conversation, the patient's speech is recorded, immediately after the call some features from the speech are extracted and analysed. The accuracy of

Table 1
Characteristics of studies investigating the effect of self-monitoring apps in BD patients.

No	Authors	Year	Treated patients	Psycho-education	Design	Outcomes
1.	Abdullah et al. (*Cornell University, USA*)[70]	2016	7	No	longitudinal	- self-assessed energy scores – correlates with sensor data including, location and communication information
2.	Bonsall et al (*University of Oxford*)	2012	23	No	longitudinal	- scores demonstrated more variability in the unstable group of patients compared with the stable group - statistical approach in characterizing mood stability and mood instability in bipolar disorder
3.	Bopp et al. (*University of Colorado, USA*)	2010	62	No	longitudinal	- the data collection allowed to examine the proportion of time patient spent in state of mania/hypomania, depression, mixed state, or euthymia, and time trends in the trajectory of symptoms - the monitoring method is readily adopted by patients with BD I and BD II
4.	Depp et al. (*University of California, USA*)[73]	2012	18	No	RCT	- significantly higher variability of self- monitored mood in the electronic self-monitoring group both within- person and between- persons - significantly higher compliance in control group
5.	Depp et al. (*University of California, USA*)	2015	82	Yes	RCT	- psychoeducation using personalized real-time intervention for mood stabilizing - feasible, satisfactory and acceptable intervention - efficacy is not proved
6.	Faurholt- Jepsen et al. (*Psychiatric Center Copenhagen, IT University of Copenhagen)*	2014b	17	No	longitudinal	- self-monitored depressive symptoms correlated significantly with HDRS-17 items score
7.	Faurholt- Jepsen et al. (*Psychiatric Center Copenhagen, IT University of Copenhagen*)	2015a	61	No	RCT	- primary outcome: no significant difference in HDRS-17 and YMRS between groups - sub-analyses: more depressive symptoms in the intervention group; fewer manic symptoms in the intervention group
8.	Faurholt- Jepsen et al. (*Psychiatric Center Copenhagen, IT University of Copenhagen*)[75]	2015c	78	No	RCT	- no significant effects on depressive as well as manic symptoms - further analyses: improving manic symptoms
9.	Faurholt- Jepsen et al. (*Psychiatric Center Copenhagen, IT University of Copenhagen*)	2015b	33	No	longitudinal	- patients with BD II had poorer outcome than patients with BD I
10.	Faurholt- Jepsen et al. (*Psychiatric Center Copenhagen, IT University of Copenhagen*)	2016b	29	No	longitudinal	- self- monitored mood and activity level correlate significantly with scores on HDRS and YMRS
11.	Hidalgo- Mazzei et al. (*Universitat de Barcelona*)	2016	51	Yes	longitudinal	- feasible, satisfactory and acceptable intervention - efficacy is not proved
12.	Vazquez-Montes et al. (*University of Oxford*)	2018	146	No	longitudinal	- tele-monitoring with control rules has the potential to predict about one-third of manic or depressive episodes before they occur in patients with bipolar disorder, at the cost of a high false positive rate - two-thirds of episodes would not be detected in advance by the control rules, but telemonitoring itself could detect all episodes at least at the moment of occurrence (usually mood episode last longer than a week)

Table 2
Characteristic of studies investigating the effect of apps collecting objective data in patients with BD.

No	Authors	Year	Treated patient	Sensors	Primary measures	Design	Findings
1	Abdullah et al. (*Cornell University, USA*)[70]	2016	7	accelerometer, microphone, GPS, MoodRhythm application	physical activity, mobility, social communication	longitudinal	social Rhythm Metric (SRM)as a clinically-validated marker of stability and rhythmicity for individuals with BD automated sensing can be used to infer the SRM score classifiers using sensor streams can predict stable and unstable states with high accuracy
2	Alvarez- Lozano et al. (*CICESE Research Center, Ensenada, Mexico*)	2014	18	Accelerometer microphone Monarca software	number of apps running app usage patterns	longitudinal	patterns of app usage vary with self- reported mood
3	Beiwinkel et al.(*Leuphana University of Lüneburg, Germany)*	2016	13	SIMBA system	physical activity, users' device activity, social communication	longitudinal	clinical symptoms are related to some objective smartphone measurement
4	Cao et al. (*University of Illinois at Chicago, USA*)	2017	20	standard Android OS keyboard accelerometer	collected metadata consisting of keypress entry time	bipolar subjects and normal controls	experimental results: 90.3% prediction accuracy on the depression score can be achieved based on session-level mobile phone typing dynamics which is typically less than one minute
6	Faurholt- Jepsen et al. (*Psychiatric Center Copenhagen, IT University of Copenhagen*)	2014	17	Accelerometer cell tower ID Monarca software	physical activity, social activity, speech duration	longitudinal	physical and social activity correlate significantly with HDRS score
8	Faurholt- Jepsen et al. (*Psychiatric Center Copenhagen, IT University of Copenhagen*)	2015a	61	Monarca 1 system	number and duration of incoming and outgoing calls, number of text messages	longitudinal	automatically collected data correlate with clinically rated depressive and manic symptoms and differ between affective state
9	Faurholt- Jepsen et al. (*Psychiatric Center Copenhagen, IT University of Copenhagen*)	2016b	29	Monarca system updated version	physical activity, social activity, phone usage, mobility	longitudinal	objective data reflect illness severity discriminate between affective state
10	Grünerbl et al. (*Embedded Intelligence, TU Kaiserslautern, Germany*)	2014	10	Accelerometer GPS	daily mobility travel patterns	longitudinal	-detection a change in mood state -less precise to detect mood state
11	Hidalgo-Mazzei et al. (*Universitat de Barcelona*)	2016	51	SIMPLe application on biological rhythms regulation	daily self-assessment test	longitudinal	positive effects of a psychoeducational smartphone application as an adjunctive to treatment as usual on BD patients' biological rhythms
12	Maxhuni et al. (CREATE-NET, Trento, Italy)	2016	5	Audio Accelerometer	phone usage, number of phone calls, duration and length of SMS and daily self-assessment questionnaires	naturalistic observation	it is possible to classify with a high confidence (> 80%) the course of mood episodes or relapse in bipolar patients
13	Osmani et al. (*CREATE-NET, Trento, Italy*)	2013	9	Accelerometer	overall activity levels	longitudinal	substantial individual variation in activity level
14	Zulueta et al. (*University of Illinois at Chicago, USA*)	2018	19	Mobile phone loaded with a customized, accelerometer keyboard	mobile phone keyboard activity	longitudinal	-evidence for the feasibility of using passively collected keyboard metadata to detect and monitor mood disturbances -mood states in BD appear to correlate with specific changes in mobile phone usage

Table 3
Characteristic of papers assessing affective episodes using voice data.

No	Authors	Year	Treated patients	Duration of monitoring	Phone calls features	Additional collected data	Accuracy for detecting the state
1	Faurholt- Jepsen et al. (*Psychiatric Center Copenhagen, IT University of Copenhagen*)	2016a	28	12 weeks	- social interaction - sound features - voice features	automatically collected objective data, self-monitored data	- manic and mixed state: AUC 0.89 - depressive state: AUC 0.78 - combining with sel-monitored and objective data accuracy increases slightly
2	Gideon et al. (*University of Michigan, USA*)	2016	15-18	29 weeks	- sound features		- manic state: AUC 0.72 ± 0.20 - for depressive state: AUC 0.75 ± 0.14
3	Grünerbl et al. (*Embedded Intelligence, TU Kaiserslautern, Germany*)	2014	10	12 weeks	- social interaction - sound features - voice features	automatically collected objective data	- state recognition: accuracy 0.80 - change detection: precision 0.97, recall 0.94
4	Karam et al. (University of Michigan, USA) [78]	2014	6	6-12 months	collected unstructured speech continuously and unobtrusively	other electronic data was not collected	- (hypo)manic state: AUC 0.81 ± 0.17 - depressive state: AUC 0.67 ± 0.18
5	Muaremi et al. (*Wearable Computing Lab, ETH Zurich, Switzerland*)	2014	12	12 weeks	- phone call statistics - social signals processing - acoustic emotion properties of the voice	automatically collected objective data, self- monitored data	- F_1 score 0.83 - not reported separately for depressive and manic states
6	Osmani et al. (*CREATE-NET, Trento, Italy*)	2015	10	12 weeks	-sound features	medical examination every three weeks, localization (GPS), accelerometer	- location: accuracy 81%, recall 81.7%, precision 80.8% - accelerometer: accuracy 72%; recall 62.9%; precision 64.8% - sound: accuracy 70%; recall 60%; precision 59%
7	Vanello et al. (*University of Pisa, Italy*)	2012	6	2 days	-sound features		- not applicable

classification of affective states based on voice features in these studies is in the range of 67%- 97%. These studies have shown the applicability of voice analysis from daily phone calls to assessing the depressive and manic episodes in BD patients in a real-life environment.

3.4. Data analysis and machine learning

The data analysis and machine learning tasks related to the phase or phase change detection in BD can be broadly divided into three groups: data preprocessing, exploratory analysis and predictive modelling.

Preprocessing largely depends on the data collection procedure and is strongly related to particular smartphone app. Several papers documents experience and best practices of mobile app development, e.g. [39–41]. Commonly stressed points include the importance of software stability, user-friendliness, the benefits of involving both patients and psychiatrists in the iterative design process and handling of missing data. Bonsall et al. [42] conducted an extensive missing values analysis and used the survival analysis to establish whether individuals were more likely to drop out from the study in one group compared with another. Moreover, Bonsall et al. [42] performed binary regressions to determine if the distribution of missing values was dependent on individual patients or group. Table 4 summarizes the characteristics of data analysis and machine learning approaches from studies using automatically collected smartphone data.

3.4.1. Exploratory analysis

Exploratory data analysis consists in summarizing the main characteristics of the observations with the goal to understand the dataset and formulate hypotheses, e.g. concerning the choice of the most clinically relevant features that can be collected by smartphones [34,37,43]. Descriptive statistics, visualization techniques and correlation analyses are among the most often used methods. Beiwinkel et al. [37] highlight the role of psychosocial variables in the illness course in BD using marginal effect plots including clinical threshold for depressive and manic symptoms.

3.4.2. Predictive modelling

Predictive modelling is most commonly stated as the determination of BD phase. This task can be formulated as classification problem in two ways: either one can discriminate between all four basic phases (euthymia, depression, mania and mixed state) or use two binary classifiers out of which one discriminates between euthymia and depression whereas the other between euthymia and mania [44,45]. The latter approach is much frequent in the literature and can be also stated as a regression problem, in which the goal is to predict the score levels on depression and mania scales [39].

Finally, there are approaches focusing on the analysis of the stability of the features with the goal to detect change irrespective of its direction [46]. Bonsall et al. [42] showed that the inter-episodic mood instability can be modelled by threshold autoregression with not normally distributed stochastic errors. Vazquez-Montes et al. [47] provided the first application in this field of control charts, a well-known tool originating from statistical process control, for monitoring mood instability of bipolar patients. Telemonitoring with control rules has the potential to predict about one-third of manic or depressive episodes before they occur [47]. Kaczmarek-Majer et al. [48] introduce a novel weighted model averaging approach to construct control charts and illustrate its performance in a case study aimed at detecting instability in processes describing behavioural data about smartphone usage of BD patients.

Prediction models can either be patient-dependent [39,49] or general [45,49–51]. The trade-off between personalization (bias) and the reduction of sample size (variance) can be somewhat counterbalanced by means of mixed models [37] in which some parameters are general while other patient-dependent. Personalization can be also conducted on a level of feature extraction, for instance Kaufmann et al. [52] relate

Table 4
Characteristic of data analysis and machine-learning approaches used for phase recognition and phase change detection.

No	Author	Year	Self-monitored data	Objective data	Voice data	Problem formulation	Problem from analytical perspective	Methods of data analysis and machine learning	Application name and data sources' description
1	Alvarez-Lozano et al. (*CICESE Research Center, Ensenada, Mexico*)	2014	Yes	Yes	No	Phase change detection	Exploratory and correlation analysis	- Pearson's correlations	MONARCA - electronic self-monitored data, objective smartphone data (acceleration, location) extended with data on smartphone applications usage, clinical evaluations
2	Beiwinkel et al.. (*Leuphana University of Lüneburg, Germany*)	2016	Yes	Yes	Yes	Identification of clinical relevance of smartphone data	Exploratory and correlation analysis	- random coefficient multilevel models - between patient and within patient effects - marginal effect plots	SIMBA - Self-reported daily mood, physical activity and social communication, clinical assessment every 8 weeks
3	Bonsall et al. (*University of Oxford, UK*)	2012	Yes	No	No	Modeling interepisodic mood instability	Time series modelling and analysis	- time series analysis (AR(1) and AR(2) for the self-assessment scores with and without threshold) - RMSE to evaluate quality of model fitting. - analysis of missing values - descriptive statistics	weekly self-rated scores: QIDS-SR, (initially also ASRM, STAI), clinical assessment not described
4	Bopp et al. (*University of Colorado, USA*)	2010	Yes	No	No	Identification of clinical relevance of smartphone data	Exploratory analysis	Descriptive statistics for BDI and BDII	OXTEXT - Text messaging weekly self-rated scores: QIDS-SR (quick inventory of depressive symptomatology) and ASRM (Altman self-rating mania scale)
5	Cao et al. (*University of Illinois at Chicago, USA*)	2017	Yes	No	No	Phase detection	Prediction of mood scores	A two-stage late fusion approach for modeling the multi-view time series data (recurrent neural network, multi-view/factorization machines)	DeepMood - a custom keyboard that collected metadata consisting of keypress entry time and accelerometer movement, clinical assessment weekly
6	Depp et al. (*University of California, USA*)[73]	2012	Yes	No	No	Identification of validity of paper-and-pencil daily mood charts	Exploratory analysis	Generalized estimating equations procedure, with a Poisson Link Function for binary data Generalized linear models procedure Inter-Quartile range	Application name not provided. Self-reported data on a mobile phone or a standard pencil-and-paper mood chart
7	Depp et al. (*University of California, USA*)	2015	Yes	Yes	No	Identification of validity of paper-and-pencil daily mood charts	Exploratory analysis	-t-student test -within-patient standard deviation -GLM procedure, -interquartile range -Pearson correlation coefficient	Application name not provided. Self-reported data and smartphone usage data by some of the participants for comparison with the pencil-and-paper study
8	Faurholt- Jepsen et al. (*Psychiatric Center Copenhagen, IT University of Copenhagen*)	2016	Yes	Yes	Yes	Phase detection	Classification of state	- random forest - random oversampling (to diminish the problem of imbalanced classes) - user-dependent models vs user-independent models	MONARCA - electronic self-monitored data, objective smartphone data (Acceleration, location), voice features, clinical assessment every 4 days
10	Gideon et al. (*University of Michigan, USA*)	2016	No	No	Yes	Phase detection	Classification of state, preprocessing of voice features	- classifiers (support vector machines with linear and RBF kernels) - two binary classification tasks: depression vs euthymia and mania vs euthymia	PRIORI app (voice features) and weekly clinician calls
11	Grünerbl et al. (*Embedded Intelligence, TU Kaiserslautern, Germany*)	2015	Yes	Yes	Yes	Phase change detection	Classification of state change	-selected classifiers (Naive Bayes, kNN, j48 search tree) from Weka -weighted FUSION concatenation (self-designed method for state change recognition)	MONARCA - electronic self-monitored data, objective smartphone data (Acceleration, location), voice features, clinical assessment weekly
12	Guidi et al. (*University of Pisa, Italy*)	2017	No	No	Yes	Phase detection	Classification of state compared to euthymia state	- spectral analysis - Mann–Whitney U-test	PSYCHE – voice recordings of reading neutral texts, clinical assessment during reading sessions
13		2017	Yes	Yes	No				

(*continued on next page*)

Table 4 (*continued*)

No	Author	Year	Self-monitored data	Objective data	Voice data	Problem formulation	Problem from analytical perspective	Methods of data analysis and machine learning	Application name and data sources' description
	Hidalgo-Mazzei et al. (*Universitat de Barcelona*)					Evaluate the potential of smartphone app to regulate biological rhythms	Exploratory and correlation analysis	- paired t-test analyses - Wilcoxon signed-rank test - Pearson´s correlations - multiple regression - ANCOVA analyses	SIMPLe – electronic self-monitored data, objective smartphone data (Acceleration, location)
14	Karam et al. (University of Michigan, USA)	2014	No	No	Yes	Phase detection	Classification	- SVMs with linear and RBF kernels - selection of acoustic features - leave-one-patient-out cross-validation	Name not provided – weekly calls with a trained clinician
15	Maxhuni et al. (*CREATE-NET, Trento, Italy*)	2016	Yes	Yes	Yes	Phase detection	Classification	- Seven classifiers (C4.5, Random Forest, SVM, Naive Bayes, k-NN, AdaBoost, Bagging) from Weka taken with their default parameters - Feature selection	MONARCA - Daily motor activity taken during phone calls, daily self-assessment (YMRS, HAMD), clinical assessment every 3 weeks
16	Muaremi et al. (*Wearable Computing Lab, ETH Zurich, Switzerland*)	2014	Yes	No	Yes	Phase detection	Classification	- random forest (RF), SVM, Logit, NN (only RF results were reported, because they were the best) - evaluation with F1 scores - leave-one-patient CV	MONARCA - electronic self-monitored data (used supplementarily), voice data, clinical assessment in hospital every 3 weeks using standardized psychological test scales; diagnosis valid for 7 days before and 2 after conduction
17	Osmani et al. (*CREATE-NET, Trento, Italy*)	2013	Yes	Yes	No	Phase change detection	Exploratory and correlation analysis	- Pearson's correlations	MONARCA - electronic self-monitored data, objective smartphone data (Acceleration, location)
18	Vazquez-Montes et al. (*University of Oxford*)	2018	Yes	No	No	Mood stability	Monitoring process stability	-statistical process control (control charts with Shewhart control rules) -general vs user-dependent models	OXTEXT - Telemonitoring of weekly rates of Altman Self-Rating Mania Scale (ASRM) and the Quick Inventory of Depressive Symptomatology—Self-Report (QIDS)

*Further details about each study (like classification accuracy) are described in Tables 1, 2 or 3 depending on the considered data.

the sleep time to the average sleep time of a given person. Generally, it seems that BD patients have a preference towards personalized, flexible applications [41,53].

Several papers consider the phase classification problem [49,45,46,54]). The following classifiers are considered: support vector machines, naive Bayes, *k* nearest neighbours, decision trees (j48 and C4.5) and ensemble methods such as random forest, AdaBoost, or bagging. However, there is no clear evidence about the superiority of any of the classifier for the BD-related data.

Another issue is the extraction and engineering of features. For instance, the number of text messages sent daily correlate with the overall activity of the patient but finer time granularity is required to analyse diurnal patterns [54]. Muaremi et al. [50] analysed voice data and recommend the average speaking length, the mean harmonics-to-noise ratio, the number of short turns/utterances, the standard deviation of the fundamental frequency (F0), and the maximum daily phone call length. Guidi et al. [55,56] showed the significance of spectral statistics derived from the fundamental frequency contours as well as prosodic features such as the total duration of pauses, the duration of long pauses and speaking rate. Grünerbl et al. [46] reported the usefulness of satellite navigation data and smartphone's acceleration.

4. Discussion

Mobile application as a monitoring tool allows for continuous assessment in naturalistic settings. Smartphones are capable of capturing multiple dimension of human behaviour, especially social activity – a key factor in BD. Multitude of complex data can be delivered through either active monitoring (self-assessment), passive monitoring of various activities, mobility or phone call statistics.

4.1. Smartphone-based data

There are many studies examining the role of self-reporting combined with psychoeducation or cognitive-behavioural therapy for major depressive disorder. Their outcomes are quite promising, showing, that such interventions lead to reductions in depressive symptoms [57,58]. However, self-reporting requires direct participation of the patient. The necessity to respond to daily surveys can be onerous for patients. Other problem is also a decreased illness insight resulting in less reliable self-identification of mood state by patients, especially in hypomanic/manic phases [34]. This motivates the search for more objective and patient-independent methods to monitor the mood shifts in patients with BD. Analysed studies have shown, that objective data are able to detect phase change more accurately than self-assessment data. Also, combining data from self-assessment with objective smartphone data on behavioural activities and voice features can increase the accuracy. Typically, a dedicated application runs in background and can gather data through smartphone's sensors [59]. Thus, it does not require any active participation of the patient and the aspect of decreased illness insight and inaccuracy of data is eliminated. The results of current research indicate that a wide range of smartphones' objective parameters could serve as a markers in monitoring of illness activity. In particular, it seems that data such as phone call statistics, speech features, phone usage, physical activity, mobility and physiological data are very promising in recognizing and predicting mood state. Faurholt-Jepsen et al. [34] showed that, the automatically collected objective data were sufficient to discriminate between affective states (asymptomatic versus moderate to severe depression or mania), suggesting that it may be a biomarker for the longitudinal monitoring of illness activity in patients with BD. This result is very promising because such electronic monitoring in patient's everyday life could enable the detection of depressive and manic episodes at their outset. Moreover Grünerbl et al. [35], in a study assessing social interaction, physical motion and travel patterns, argue that change detection is more important than the recognition of a particular state.

However, it is worth noticing, that also behavioural objective data can be very patient-dependent (depending on the individual characteristics of the patient, not just the disease itself). Therefore the search went even further into entirely patient-independent markers, such as voice data or the parameters of the autonomic system. The results of analyzed studies indicate that voice data from everyday phone calls could be a valid marker of a mood state and seem promising in monitoring the illness activity [51]. However, at this moment, as Faurholt-Jepsen et al. (2018) note, it is too early to specify which of the objective smartphone data have the strongest correlation with the clinically assessed depressive or manic symptoms. Also, further studies are necessary before considering voice features as a marker of early detection of prodromal symptoms of phase change in BD patients.

4.2. Machine learning challenges and limitations in reviewed studies

The state-of-the art solutions have approximately accuracy ranging from 67% to 97%, however, the sample sizes and the metrics used to report the accuracy are different between the reviewed works. Most papers using acoustic data present results of a limited sample size (6-28 patients). Data sharing could largely improve the validity of the results, however, it is prevented by sensitive and private character or voice, localization and personal information. Anonymization techniques, such as scrambling of the speech signal [50] or removing unvoiced frames [26] are necessary to overcome this issue.

As observed in the analysed papers, that there is no common standard applied when reporting results of phase change classification. Several measures are used to describe its quality: the area under curve (AUC) [45,51], accuracy, precision and recall (e.g., [54,80]), F_1 score (e.g., [50]). Although accuracy defined as the fraction of correctly predicted cases seems the easiest and most appealing, it might be misleading if the test sample is unbalanced. In particular, a trivial rule always predicting the most common state has an accuracy equal to the frequency of this category.

Consequently, additional measures are recommended. In particular, false-positive errors are quantified by precision while false-negative by recall (or sensitivity). A harmonic mean between these quantities, called F_1 score, is highly recommended although seldomly used in the reviewed papers. Most classifiers have parameters (such as thresholds) which allow adjusting the emphasis put on precision or recall. The trade-off between the true positive and false positive rates is often represented by plotting them for different thresholds leading to a so-called receiver operator curve. The Area Under the Curve (AUC) describes the probability that a classifier ranks a random positive instance higher than a random negative instance [60].This measure is relatively robust against class imbalance. AUC, as well as F1 score, are both recommended in other severe psychiatric conditions such as major depression or suicidal risk assessment [26].

Furthermore, the current state of the art seem to lack clear guidelines about the ground-truth before and after the psychiatric assessment and its selection may significantly influence the results. Grünerbl et al. [46] consider the period of 7 days before and 2 days after the examination (and phone interview) as ground-truth for analyses.

Predictive modelling is most commonly stated as phase classification rather than phase change classification or regression problem. The pool of machine learning techniques is broad and only a small part of it (e.g., random forests, Naïve Bayes) have been verified in the context of supporting BD patients. In particular, application of deep neural networks with recurrent architecture seems to be a promising direction [61], as these methods proved very effective in speech recognition and acoustic modelling tasks [62]. During the Audio/Visual Emotion Challenge 2018 a special session was dedicated to the analysis of recordings (covering audio, video and text features) of BD patients. Various models were compared on the same dataset under the same experimental and evaluation conditions which allowed for benchmarking the performance of different algorithms. To benefit from all

type of data, multi-modal fusion algorithms are necessary [63].

4.3. Smartphone implementation challenges

The diversity of smartphone hardware is another challenge as varying recording quality requires appropriate preprocessing [45]. The fact that the two most widespread mobile operating systems (Android and iOS) are only partially compatible increases the development costs. The published studies used different applications, clinical rating scales, methodology, different definitions of affective states which makes it difficult to compare results across studies and to draw definite conclusions. Further, well-designed studies, with a larger study group are needed to confirm the validity of smartphone-based app to monitor and predict the phase change in BD.

4.4. Future of smartphone-based bipolar disorder monitoring

Smartphone-based diagnostic methods are relatively new, therefore many monitoring apps have some shortcomings. Both, the technical side of privacy ensuring, as well as substantive content tend to be imperfect. For instance, less than a quarter of apps for monitoring BD provides a privacy policy and only a few of them are in line with evidence-based medicine [13]. Transition of the research results to technology is hindered by high heterogeneity of BD symptoms and the lack of reproducibility. Several common problems occurring in mHealth apps are detailed by Huckvale et al. [64].

The ultimate goal is the development of an effective tool not only to monitor, but primarily to predict and detect the phase change in BD. Smartphone applications along with structural imaging techniques, gene testing, drug development, pattern recognition, and electroencephalography (EEG)-based diagnosis can be part of the artificial intelligence systems for managing mood disorders [65].

The early detection of subthreshold symptoms of mood episode recurrence, as well as prodromes preceding an initial episode (e.g. among individuals at high risk of conversion to bipolar disorder) could be of vital importance. Nevertheless, further research is needed including larger samples of patients, longer study periods and addressing confounding factors in the analyses to develop effective tools for monitoring and predicting the phase change. Mobile apps and machine learning technologies could also be useful in the early intervention in the course of other chronic, recurrent mental disorders such as borderline personality disorder, recurrent depression, schizophrenia [66–68], as well as for early detection of suicide risk in various mental conditions [69].

5. Conclusions

Smartphone-based monitoring has clear potential for facilitating timely and contextual care delivery for patients with bipolar disorder. Mobile apps using permanent, real-time monitoring of illness activity that could detect the phase change, would be very helpful to patients and clinicians. This could improve the diagnosis and provide options for early intervention on prodromal symptoms between outpatient visits in BD patients.

Author statement

A. Z. Antosik-Wójcińska, M. Chojnacka and M. Dominiak were responsible for the study concept and design, performed the literature search and analysis concerning medical part of the study and drafted the manuscript. Ł. Święcicki and M. Dominiak assisted with data analysis and interpretation of findings. K. Kaczmarek-Majer, K. R. Opara, W. Radziszewska and A. Olwert were responsible for the study design, performed the literature search and analysis concerning technical part of the study and drafted the manuscript. Ł. Święcicki, M. Dominiak, K. Kaczmarek-Majer and K. R. Opara, provided critical revision of the manuscript for important intellectual content. All authors critically reviewed content and approved final version for publication. This manuscript describes original work and has not been published elsewhere and is not under consideration by another journal.

Declaration of Competing Interest

The authors report no conflict of interest.

Summary points

What was already known on the topic?

- Smartphones are capable of capturing multiple dimension of human behaviour and activity – a key factors in chronic, recurrent mental disorders, especially in bipolar disorder (BD)
- Mobile applications allow for continuous monitoring of illness activity in BD patients in naturalistic settings
- Wide range of smartphones' subjective (self-assessment) and objective parameters (mobility, physical activity, phone call statistics, voice features) could serve as a markers in monitoring of BD activity
- Data analysis and machine learning aspects are gaining attention for supporting the smartphone-based BD systems, however, only a few have been verified in the context of supporting BD patients

What this study added to our knowledge?

- Objective data, including voice features, are able to detect and predict phase change in BD more accurately than self-assessment data
- There is still need for common standards and clear guidelines about metrics to be applied when reporting results of the BD phase and phase change predictive modelling
- Various machine learning approaches (non-linear autoregressive models, mixed effects models, neural networks) were applied in this field
- Application of deep neural networks with recurrent architecture seems to be a promising direction, as these methods proved very effective in speech recognition and acoustic modelling tasks
- The state-of-the art solutions report accuracy ranging from 67% and higher for the BD phase classification, however, it seems too early to suggest the higher adequacy of any of the approach (regression or classification) or any particular classification or regression method

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References

[1] F. Catala-Lopez, R. Genova-Maleras, E. Vieta, R. Tabares-Seisdedos, The icreasing burden of mental and neurological disorders, Eur Neuropsychopharmacol 23 (2013) 1337–1339.
[2] I. Grande, M. Berk, B. Birmaher, E. Vieta, Bipolar disorder, Lancet 387 (2016) 1561–1572.
[3] R. Lee, Bipolar Disorders: Mixed states, rapid cycling and atypical forms, Journal of Mental Health 17 (1) (2008) 123, https://doi.org/10.1080/09638230701530283.
[4] M.A. Namjoshi, D.P. Buesching, A review of the health-related quality of life literature in bipolar disorder, Qual Life Res 10 (2001) 105–115.
[5] C.E. Begley, J.F. Annegers, A.C. Swann, C. Lewis, S. Coan, W.B. Schnapp, L. Bryant-Comstock, The lifetime cost of bipolar disorder in the US: an estimate for new cases in 1998, Pharmacoeconomics 19 (2001) 483–495.

[6] G.M. Goodwin, J.R. Geddes, Latest maintenance data on lithium in bipolar disorder, Eur Neuropsychopharmacol 13 (suppl. 2) (2003) 51–55.
[7] E. Vieta, E. Salagre, I. Grande, A.F. Carvalho, B.S. Fernandes, M. Berk, et al., Early Intervention in Bipolar Disorder 5 (175) (2018) 411–426.
[8] C.A. Crisan, Lack of Insight in Bipolar Disorder: The Impact on Treatment Adherence, Adverse Clinical Outcomes and Quality of Life, Psychotic Disorders – an update, Intechopen, 2017.
[9] L.N. Yatham, S.H. Kennedy, S.V. Parikh, A. Schaffer, D.J. Bond, B.N. Frey, et al., Canadian Network for Mood and Anxiety Treatments (CANMAT) and International Society for Bipolar Disorders (ISBD) 2018 guidelines for the management of patients with bipolar disorder, Bipolar Disorders 20 (2) (2018) 97–170.
[10] D.I. Velligan, P.J. Weiden, M. Sajatovic, J. Scott, D. Carpenter, R. Ross, J.P. Docherty, Strategies for addressing adherence problems in patients with serious and persistent mental illness: recommendations from the expert consensus guidelines, J Psychiatr Pract. 16 (5) (2010) 306–324.
[11] C. Schneck, Bipolar Phenomenology: Have We Learned All We Can Learn? Am J of Psychiatry 168 (1) (2011) 4–6.
[12] M.W. Zhang, R.C. Ho, Smartphone application for multi-phasic interventional trials in psychiatry: Technical design of a smart server, Technol Health Care 25 (2) (2017) 373–375.
[13] J. Nicholas, M.E. Larsen, J. Proudfoot, H. Christensen, Mobile apps for bipolar disorder: a systematic review of features and content quality, Journal of Medical Internet Research 17 (8) (2015).
[14] S. Fealy, S. Chan, O. Wynne, E. Dowse, L. Ebert, R. Ho, et al., The Support for New Mums Project: A protocol for a pilot randomized controlled trial designed to test a postnatal psychoeducation smartphone application, J Adv Nurs. 75 (6) (2019) 1347–1359.
[15] M.W. Zhang, R.C. Ho, Tapping onto the Potential of Smartphone Applications for Psycho-Education and Early Intervention in Addictions, Front Psychiatry 7 (2016) 40.
[16] M.W. Zhang, S. Chan, O. Wynne, S. Jeong, S. Hunter, A. Wilson, R.C. Ho, Conceptualization of an evidence-based smartphone innovation for caregivers and persons living with dementia, Technol Health Care 24 (5) (2016) 769–773.
[17] M. Faurholt-Jepsen, M. Frost, M. Vinberg, E.M. Christensen, J.E. Bardram, L.V. Kessing, Smartphone data as objective measures of bipolar disorder symptoms, Psychiatry Research 217 (2014) 124–127.
[18] M. Faurholt- Jepsen, C. Ritz, M. Frost, R.L. Mikkelsen, E.M. Christensen, J. Bardram, L.V. Kessing, Mood instability in bipolar disorder type I versus type II- continuous daily electronic self- monitoring of illness activity using smartphones, J Affect Disord 186 (2015) 342–349.
[19] C.A. Depp, J. Ceglowski, V.C. Wang, F. Yaghouti, B.T. Mausbach, W.K. Thompson, E.L. Granholm, Augmenting psychoeducation with a mobile intervention for bipolar disorder: A randomized controlled trial, J Affect Disord 174 (2015) 23–30.
[20] L.O. Scharer, V. Hartweg, G. Valerius, M. Graf, M. Hoern, C. … Biedermann, J. Walden, Life charts on a palmtop computer: first results of a feasibility study with an electronic diary for bipolar patients, Bipolar Disord 4 (Suppl. 1) (2002) 107–108.
[21] J.M. Bopp, D.J. Miklowitz, G.M. Goodwin, W. Stevens, J.M. Rendell, J.R. Geddes, The longitudinal course of bipolar disorder as revealed through weekly text messaging: a feasibility study, Bipolar Disord 12 (3) (2010) 327–334.
[22] J.F. Greden, B.J. Carrol, Decrease in speech pause times with treatment of endogenous depression, Biol Psychiatry 15 (1980) 575–587.
[23] M. Alpert, E.R. Pouget, R.R. Silva, Reflection of depression in acoustic measures of the patient's speech, J Affect Disord 66 (2001) 59–69.
[24] J.C. Mundt, A.P. Vogel, D.E. Feltner, W.R. Lenderking, Vocal acoustic biomarkers of depression severity and treatment response, Biol. Psych 72 (2012) 580–587.
[25] N. Vanello, A. Guidi, C. Gentili, S. Werner, G. Bertschy, G. Valenza, E.P. Scilingo, Speech analysis for mood state characterization in bipolar patients, Engineering in Medicine and Biology Society (EMBC), 2012 Annual International Conference of the IEEE, IEEE, 2012, pp. 2104–2107.
[26] N. Cummins, S. Scherer, J. Krajewski, S. Schnieder, J. Epps, T.F. Quatieri, A review of depression and suicide risk assessment using speech analysis, Speech Communication 71 (2015) 10–49.
[27] M. Faurholt-Jepsen, M. Bauer, L.V. Kessing, Smartphone-based objective monitoring in bipolar disorder: status and considerations, Int J Bipolar Disord 6 (2018) 6.
[28] A. Rajagopalan, P. Shah, M.W. Zhang, R.C. Ho, Digital platforms in theassessment and monitoring of patients with bipolar disorder, Brain Sciences 7 (11) (2017) Nov 12.
[29] H. Daus, N. Kislicyn, S. Heuer, M. Backenstrass, Disease management apps and technical assistance systems for bipolar disorder: Investigating the patients' point of view, Journal of Affective Disorders 15 (229) (2018) 351–357 Mar.
[30] M. Faurholt-Jepsen, M. Vinberg, M. Frost, E.M. Christensen, J.E. Bardram, L.V. Kessing, Smartphone data as an electronic biomarker of illness activity in bipolar disorder, Bipolar Disord 17 (2015) 715–728.
[31] S. Monteith, T. Glenn, J. Geddes, P.C. Whybrow, M. Bauer, Big data for bipolar disorder, International Journal of Bipolar Disorders 4 (2016) 10.
[32] D. Hidalgo-Mazzei, A. Mateu, M. Reinares, A. Murru, C. Bonnin, C. Varo, et al., Psychoeducation in bipolar disorder with SIMPLe smartphone application: feasibility, acceptability and satisfaction, J Affect Disord 200 (2016) 58–66.
[33] D. Hidalgo-Mazzei, M. Reinares, A. Mateu, M.F. Juruena, A.H. Young, V. Perez-Sola, et al., Is a SIMPLe smartphone application capable of improving biological rhythms in bipolar disorder? Journal of affective disorders 223 (2017) 10–16.
[34] M. Faurholt-Jepsen, M. Vinberg, M. Frost, S. Debel, E.M. Christensen, J.E. Bardram, L.V. Kessing, Behavioral activities collected through smartphones and the association with illness activity in bipolar disorder, Int J Methods Psychiatr Res 25 (2016) 309–323.
[35] A. Grünerbl, V. Osmani, G. Bahle, J.C. Carrasco, S. Oehler, O. Mayora, et al., Using smart phone mobility traces for the diagnosis of depressive and manic episodes in bipolar patients, Augmented Human (AH) (2014).
[36] V. Osmani, A. Maxhuni, A. Grünerbl, P. Lukowicz, C. Haring, O. Mayora, Monitoring activity of patients with bipolar disorder using smart phones, Conference Paper, Dec (2013), https://doi.org/10.1145/2536853.2536882.
[37] T. Beiwinkel, S. Kindermann, C. Kerl, J. Moock, G. Barbian, W. Roessler, Using smartphones to monitor bipolar disorder symptoms: a pilot study, JMIR Ment Health 3 (1) (2016) e2.
[38] J. Zulueta, A. Piscitello, M. Rasic, R. Easter, P. Babu, S.A. Langenecker, et al., Predicting Mood Disturbance Severity with Mobile Phone Keystroke Metadata: A BiAffect Digital Phenotyping Study, J Med Internet Res 20 (7) (2018) e241.
[39] M. Matthews, S. Abdullah, E. Murnane, S. Voida, T. Choudhury, G. Gay, E. Frank, Development and evaluation of a smartphone-based measure of social rhythms for bipolar disorder, Assessment 23 (4) (2016) 472–483.
[40] A.M. Bauer, S. Hodsdon, S. Hunter, Y. Choi, J. Bechtel, J.C. Fortney, Lessons from the deployment of the SPIRIT app to support collaborative care for rural patients with complex psychiatric conditions, Proceedings of the 2017 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2017 ACM International Symposium on Wearable Computers, ACM, 2017, pp. 772–780.
[41] J.E. Bardram, M. Frost, K. Szántó, M. Faurholt-Jepsen, M. Vinberg, L.V. Kessing, Designing mobile health technology for bipolar disorder: a field trial of the monarca system, Proceedings of the SIGCHI conference on human factors in computing systems, ACM, 2013, pp. 2627–2636.
[42] M.B. Bonsall, S.M.A. Swallace-Hadrill, J.R. Geddes, G.M. Goodwin, E.A. Holmes, Nonlinear time-series approaches in characterizing mood stability and mood instability in bipolar disorder, Proceedings of the Royal Society of London, Series B: Biological Sciences 279 (2012) 916–924.
[43] J. Alvarez-Lozano, V. Osmani, Mayora, M.O. Frost, J. Bardram, M. Faurholt-Jepsen, L.V. Kessing, Tell me your apps and I will tell you your mood: correlation of apps usage with bipolar disorder state, ACM Proceedings of the 7th international conference on pervasive technologies related to assistive environments, New York: ACM, 2014, p. 19.
[44] B. Cao, L. Zheng, C. Zhang, P.S. Yu, A. Piscitello, J. Zulueta, A.D. Leow, DeepMood: Modeling Mobile Phone Typing Dynamics for Mood Detection, Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, ACM, 2017, pp. 747–755.
[45] J. Gideon, E.M. Provost, M. McInnis, Mood state prediction from speech of varying acoustic quality for individuals with bipolar disorder, Proc IEEE Int Conf Acoust Speech Signal Process. 2016 Mar (2016) 2359–2363.
[46] A. Grünerbl, A. Muaremi, V. Osmani, G. Bahle, S. Oehler, G. Tröster, P. Lukowicz, Smartphone-based recognition of states and state changes in bipolar disorder patients, IEEE Journal of Biomedical and Health Informatics 19 (1) (2015) 140–148.
[47] M. Vazquez-Montes, R. Stevens, R. Perera, K. Saunders, J.R. Geddes, Control charts for monitoring mood stability as a predictor of severe episodes in patients with bipolar disorder, International Journal of Bipolar Disorder 6 (7) (2018).
[48] Kaczmarek-Majer, K. Hryniewicz, O., Opara, K., Radziszewska, W., Olwert, A., Owsiński, J., Zadrożny, S. (2019). Control Disorder. In: Destercke S., Denoeux T., Charts Designed Using Model Averaging Approach for Phase Change Detection in Bipolar Gil M., Grzegorzewski P., Hryniewicz O. (eds) Uncertainty Modelling in Data Science. SMPS 2018. Advances in Intelligent Systems and Computing vol 832. Springer. (2019).
[49] A. Maxhuni, A. Munoz-Melendez, V. Osmani, H. Perez, O. Mayora, E.F. Morales, Classification of bipolar disorder episodes based on anlysis of voice and motor activity of patients, Pervasive and Mobile Computing 31 (2016) 50–66.
[50] A. Muaremi, F. Gravenhorst, A. Grünerbl, B. Arnrich, G. Troster, Assessing bipolar episodes using speech cues derived from phone calls, Interntional Symposium on Pervasive Computing Paradigms for Mental Health (mindCare) (2014) 103–114.
[51] M. Faurholt-Jepsen, J. Busk, M. Frost, M. Vinberg, E.M. Christensen, O. … Winther, L.V. Kessing, Voice analysis as an objective state marker in bipolar disorder, Transl Psychiatry 19 (6) (2016) e856 Jul.
[52] C.N. Kaufmann, A. Gershon, L.T. Eyler, C.A. Depp, Clinical significance of mobile health assessed sleep duration and variability in bipolar disorder, Journal of Psychiatric Research 81 (2016) 152–159.
[53] K.E.A. Saunders, A.C. Bilderbeck, P. Panchal, L.Z. Atkinson, J.R. Geddes, G.M. Goodwin, Experiences of remote mood and activity monitoring in bipolar disorder: a qualitative study, European Psychiatry 41 (2017) 115–121.
[54] V. Osmani, Smartphones in mental health: detecting depressive and manic episodes, IEEE Pervasive Computing 14 (3) (2015) 10–13.
[55] A. Guidi, S. Salvi, M. Ottaviano, C. Gentili, G. Bertschy, D. de Rossi, et al., Smartphone Application for the Analysis of Prosodic Features in Running Speech with a Focus on Bipolar Disorders: System Performance Evaluation and Case Study, Sensor (Basel) 15 (11) (2015) 28070–28087 Nov.
[56] A. Guidi, J. Schoentgen, G. Bertschy, C. Gentili, E.P. Scilingo, N. Vanello, Features of vocal frequency contour and speech rhythm in bipolar disorder, Biomedical Signal Processing and Control 37 (2017) 23–31.
[57] S.D. Kauer, S.C. Reid, A. Crooke, Self-monitoring using mobile phones in the early stages of adolescent depression: randomized controlled trial, J Med Internet Res 14 (3) (2012) e67 Jun 25.
[58] S. Watts, A. Mackenzie, C. Thomas, A. Griskaitis, CBT for depression: a pilot RCT comparing mobile phone vs computer, BMC Psychiatry (2013) 49.
[59] F. Gravenhorst, A. Muaremi, J. Bardram, A. Grünerbl, O. Mayora, G. Wurzer, G. Tröster, Mobile phones as medical devices in mental disorder treatment: an overview, Personal and Ubiquitous Computing 19 (2) (2015) 335–353.
[60] T. Fawcett, An introduction to ROC analysis, Pattern recognition letters 27 (8)

(2006) 861–874.
[61] Z. Du, W. Li, D. Huang, Y. Wang, Bipolar Disorder Recognition via Multi-scale Discriminative Audio Temporal Representation, Proceedings of the 2018 on Audio/Visual Emotion Challenge and Workshop, ACM, 2018, pp. 23–30.
[62] W. Liu, Z. Wang, X. Liu, N. Zeng, Y. Liu, F.E. Alsaadi, A survey of deep neural network architectures and their applications, Neurocomputing 234 (2017) 11–26.
[63] X. Xing, B. Cai, Y. Zhao, S. Li, Z. He, W. Fan, Multi-modality Hierarchical Recall based on GBDTs for Bipolar Disorder Classification, Proceedings of the 2018 on Audio/Visual Emotion Challenge and Workshop, ACM, 2018, pp. 31–37.
[64] K. Huckvale, J.T. Prieto, M. Tilney, P.J. Benghozi, J. Car, Unaddressed privacy risks in accredited health and wellness apps: a cross-sectional systematic assessment, BMC Medicine 13 (1) (2015) 214.
[65] B.X. Tran, R.S. McIntyre, C.A. Latkin, et al., The Current Research Landscape on the Artificial Intelligence Application in the Management of Depressive Disorders: A Bibliometric Analysis, Int J Environ Res Public Health 16 (12) (2019) 2150.
[66] A. Tsanas, K.E. Saunders, A.C. Bilderbeck, N. Palmius, M. Osipov, G.D. Clifford, G.M. Goodwin, M. De Vos, Daily longitudinal self-monitoring of mood variability in bipolar disorder and borderline personality disorder, J Affect Disord 15 (205) (2016) 225–233, https://doi.org/10.1016/j.jad.2016.06.065.
[67] S. Jimenez-Serrano, S. Tortajada, J.M. Garcia-Gomez, A Mobile Health Application to Predict Postpartum Depression Based on Machine Learning, Telemed J E Health (7) (2015) 567–574, https://doi.org/10.1089/tmj.2014.0113 2015 Jul 21.
[68] S. Shin, C.W. Yeom, C. Shin, J.H. Shin, J.H. Jeong, J.U. Shin, Y.R. Lee, Activity monitoring using a mHealth device and correlations with psychopathology in patients with chronic schizophrenia, Psychiatry Res 30 (246) (2016) 712–718, https://doi.org/10.1016/j.psychres.2016.10.059 Dec.
[69] N.B. Allen, B.W. Nelson, D. Brent, R.P. Auerbach, Short-term prediction of suicidal thoughts and behaviors in adolescents: Can recent developments in technology and computational science provide a breakthrough? J Affect Disord 1 (250) (2019) 163–169, https://doi.org/10.1016/j.jad.2019.03.044.
[70] S. Abdullah, M. Matthews, E. Frank, G. Doherty, G. Gay, T. Choudhury, Automatic detection of social rhythms in bipolar disorder, J Am Med Inform Assoc 23 (3) (2016) 538–543.
[73] C.A. Depp, D.H. Kim, L.V. de Dios, V. Wang, J. Ceglowski, A pilot study of mood ratings captured by mobile phone versus paper-and-pencil mood charts in bipolar disorder, J Dual Diagn 8 (2012) 326–332.
[75] M. Faurholt-Jepsen, M. Frost, C. Ritz, E.M. Christensen, A.S. Jacoby, R.L. Mikkelsen, L.V. Kessing, Daily electronic self-monitoring in bipolar disorder using smartphones- the MONARCA I trial: a randomized, placebo-controlled, single-blind, parallel group trial, Psychological Medicine 45 (2015) 2691–2704.
[78] Z.N. Karam, E.M. Provost, S. Singh, J. Montgomery, C. Archer, G. Harrington, M.G. Mcinnis, Ecologically valid long-term mood monitoring of individuals with bipolar disorder using speech, Proc IEEE Int Conf Acoust Speech Signal Process; Article number 6854525 (2014) 4858–4862.
[79] O. Carr, K.E.A. Saunders, A. Tsanas, A.C. Bilderbeck, N. Palmius, J.R. Geddes, R. Foster, G.M. Goodwin, M. De Vos, Variability in phase and amplitude of diurnal rhythms is related to variation of mood in bipolar and borderline personality disorder, Sci. Rep. 8 (2018) 1649.
[80] A. Grünerbl, A. Muaremi, V. Osmani, G. Bahle, S. Oehler, G. Tröster, P. Lukowicz, Smartphone-based recognition of states and state changes in bipolar disorder patients, IEEE J. Biomed Health Informatics 19 (2015) 140–148.